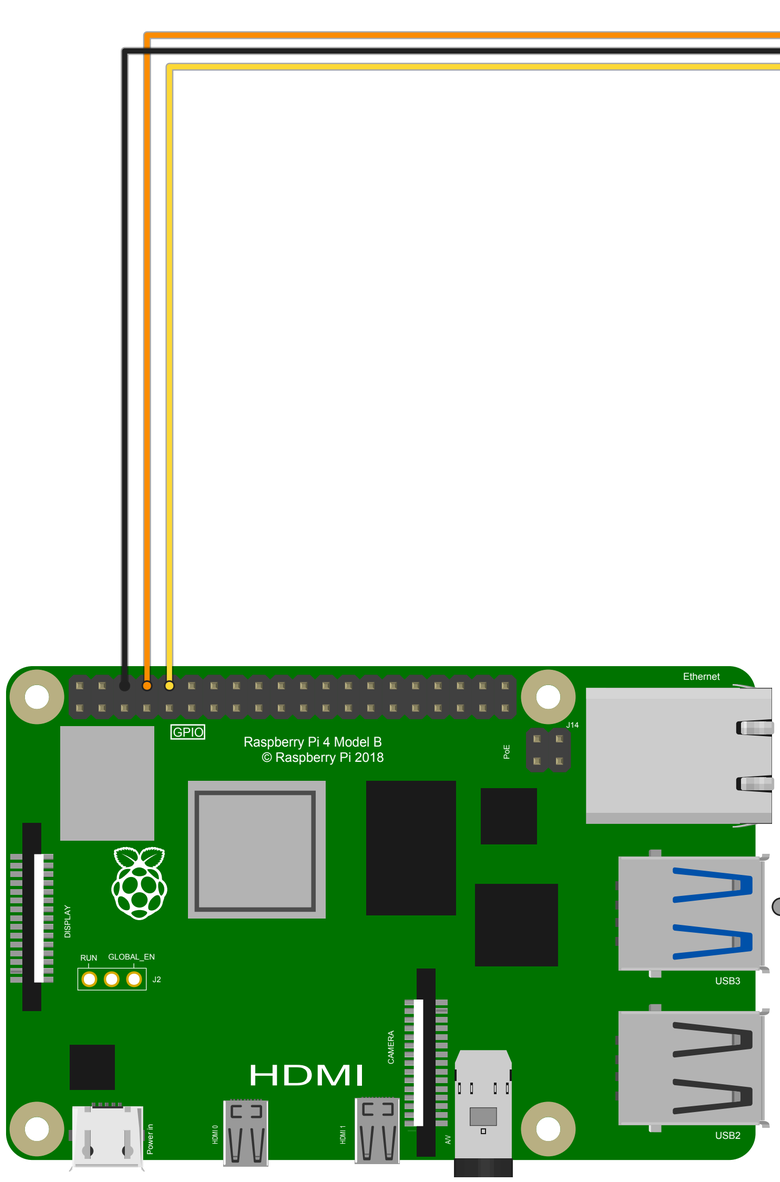
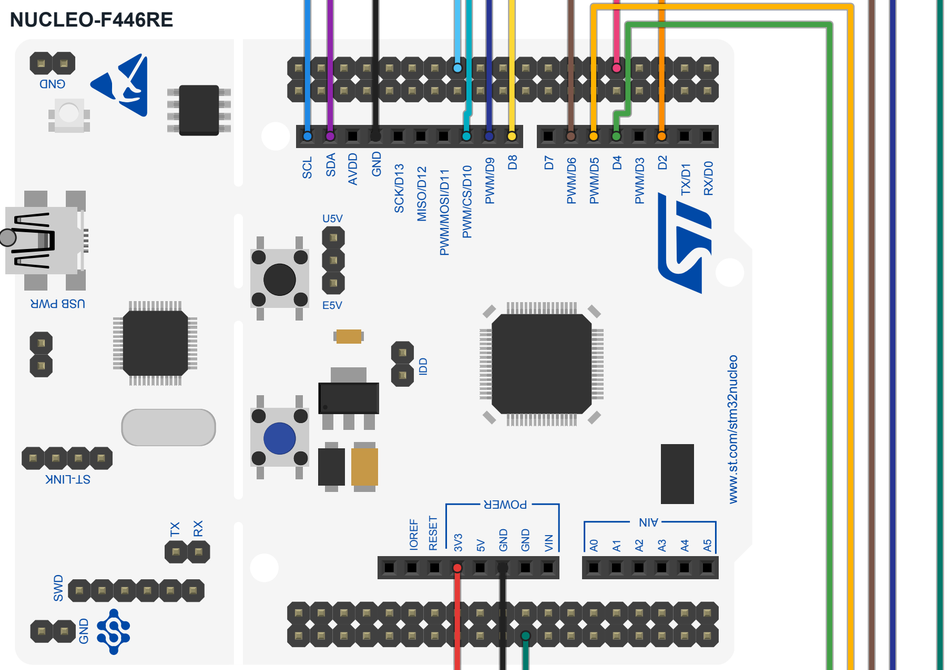
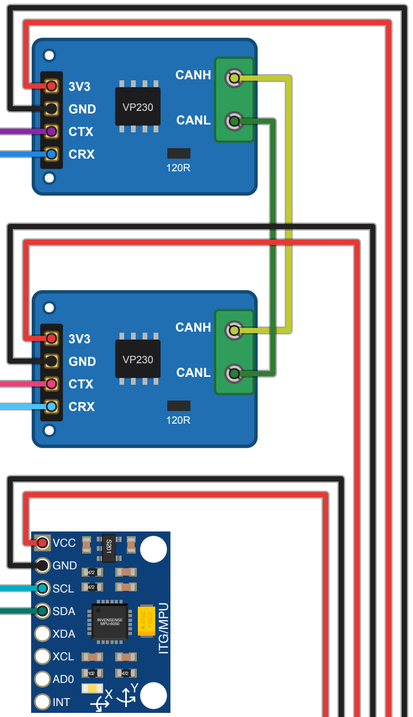
Bench wiring: Raspberry Pi 5, NUCLEO-F446RE, 2 x CAN transceiver, MPU6050, advisory LEDs



Raspberry Pi 5 (Pi 4B drawing; same 40-pin header)



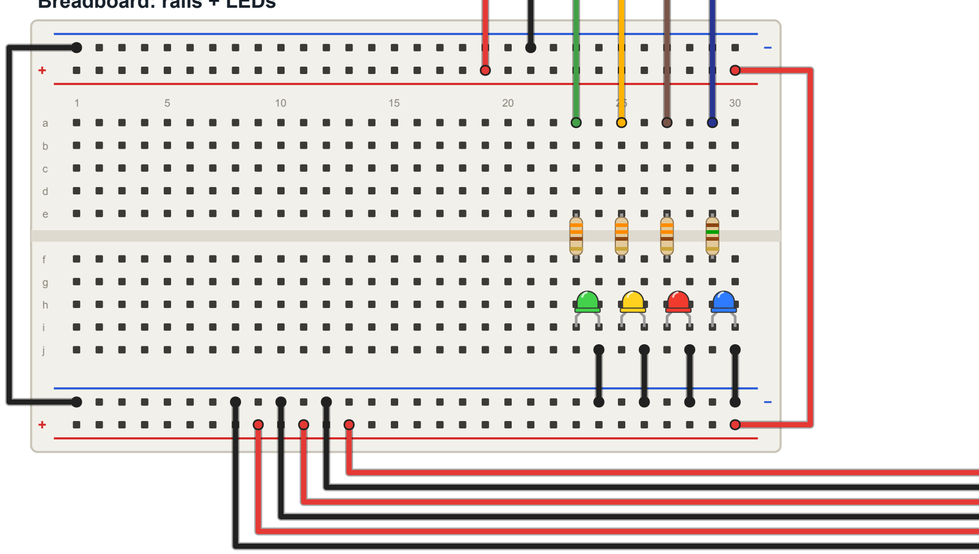


CAN #1 (SN65HVD230)

CAN bus 1 Mbit/s
120 Ω on each module
(not measured)

CAN #2 ("fake motor")

GY-521 MPU6050



Wires (pin names as on the NUCLEO silkscreen / CN10 morpho)

- Pi pin 8 GPIO14 TXD → D2 = PA10 USART1 RX
- Pi pin 10 GPIO15 RXD ← D8 = PA9 USART1 TX
- GND (Pi pin 6 → CN5 GND; rails; modules; LED cathodes)
- 3V3 (POWER header → rails → module VCC)
- USB-A → ST-LINK mini-USB (power + flashing)
- D14 = PB9 CAN1 TX → CAN #1 CTX
- D15 = PB8 CAN1 RX ← CAN #1 CRX
- CN10-30 = PB13 CAN2 TX → CAN #2 CTX
- CN10-16 = PB12 CAN2 RX ← CAN #2 CRX
- CANH CAN #1 ↔ CAN #2
- CANL CAN #1 ↔ CAN #2
- D10 = PB6 I2C1 SCL → MPU6050 SCL
- CN7-21 = PB7 I2C1 SDA ↔ MPU6050 SDA
- D4 = PB5 → 330 Ω → green LED: CLEAR (NONE)
- D5 = PB4 → 330 Ω → yellow LED: JUMP; blink = UNCERTAIN
- D6 = PB10 → 330 Ω → red LED: STOP; blink = no Pi data
- D9 = PC7 → 150 Ω → blue LED: UNCERTAIN

Board drawings: Fritzing parts library (Raspberry Pi 4B, Nucleo-64, GY-521), CC-BY-SA 3.0. CAN modules, breadboard, resistors and LEDs drawn by tools/wiring_diagram. Positions on the breadboard are illustrative; every connection matches the bench pin table.